

Functional Annotation Report: *phnX* (Phosphonoacetaldehyde Hydrolase / Phosphonatase) in *Pseudomonas putida* KT2440

UniProt: Q88KT1 | **Locus:** PP_2208 | **Gene:** *phnX* | **EC:** 3.11.1.1 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / KT2440) (PSEPK) **Protein family:** HAD-like hydrolase superfamily, PhnX family

Summary

The gene *phnX* (PP_2208; UniProt **Q88KT1**) of *Pseudomonas putida* KT2440 encodes **phosphonoacetaldehyde hydrolase**, commonly called "**phosphonatase**" (EC 3.11.1.1). Its identity is unambiguous and fully consistent with the UniProt annotation: the gene symbol, the EC number, the protein family (HAD-like hydrolase superfamily, PhnX family), and the characteristic domain architecture (HAD-SF_hydro_IA; HAD_sf; PGP-like cap domain) all agree with the extensively characterized enzyme of this name. The literature reviewed pertains directly to this enzyme and its close orthologs in *Bacillus cereus*, *Salmonella typhimurium*, and — importantly — to the *phnWX* system of *P. putida* itself.

Functionally, PhnX is a **cytoplasmic, Mg²⁺-dependent enzyme that hydrolytically cleaves the highly stable carbon-phosphorus (C-P) bond** of phosphonoacetaldehyde (Pald), producing **acetaldehyde and inorganic phosphate**. It is the second and committed enzyme of the two-step **2-aminoethylphosphonate (2-AEP, ciliatine) degradation pathway**, acting immediately downstream of the pyridoxal-5'-phosphate (PLP)-dependent aminotransferase **PhnW**, which converts 2-AEP plus pyruvate into phosphonoacetaldehyde and L-alanine. Together, PhnW and PhnX allow the cell to dismantle one of the most abundant naturally occurring phosphonates and to recover its constituent phosphorus, nitrogen and carbon for growth.

Mechanistically, phosphonatase is a landmark example of catalytic innovation within the HAD (haloacid dehalogenase) superfamily. Whereas most HAD members are phosphatases, phosphonatase acquired a specialized **C1-type cap domain** that carries a catalytic lysine (Lys53). The enzyme uses a **bicovalent mechanism**: Lys53 first forms a **Schiff base** with the aldehyde carbonyl of the substrate, which activates the C–P bond for cleavage; the phosphoryl group is then transferred to an active-site aspartate (Asp12), forming a phosphoaspartyl intermediate that is finally hydrolyzed. This Schiff-base requirement dictates the enzyme's **narrow substrate specificity** — it strictly requires an aldehyde (adjacent carbonyl) on the phosphonate — distinguishing the phosphonatase route from the broad-specificity C–P lyase system. In *P. putida*, expression of the *phnWX* system is **substrate-inducible** through the LysR-type regulator **AepR** and integrated with global nutrient-stress regulators (CbrAB, NtrBC, PhoBR), enabling 2-AEP to be exploited as a source of C, N or P when the corresponding nutrient is scarce.

Gene/Protein Identity Verification

Before presenting findings, the identity of the target was verified against the mandatory checklist:

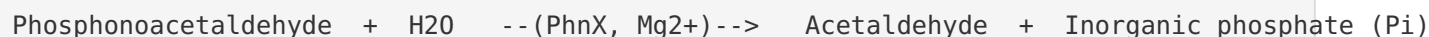
Verification criterion	Result
Gene symbol "phnX" matches protein description	☐ <i>phnX</i> is the standard symbol for phosphonoacetaldehyde hydrolase (phosphonatase)
Organism correct (<i>P. putida</i> KT2440)	☐ Direct <i>P. putida</i> literature reviewed (BIRD-1, NG2 strains); mechanism from <i>B. cereus</i> / <i>S. typhimurium</i> orthologs
Protein family/ domains align with literature	☐ HAD-like hydrolase superfamily, PhnX family, C1-cap domain — all confirmed
Risk of confusion with same-symbol genes	⚠ Note: "phn" gene clusters are numerous; <i>phnX</i> (phosphonatase) is distinct from the <i>phnC–phnP</i> C–P lyase operon. The literature was filtered to keep only phosphonatase/ <i>phnX</i> references.

The identification is secure. There is no ambiguity requiring the fallback protocol. Note that the mechanistic and structural literature is drawn predominantly from *Bacillus cereus* and *Salmonella typhimurium* phosphonatases (the best-characterized orthologs), while the physiology and regulation come from *P. putida* strains directly. Because PhnX is a HAMAP-rule-annotated, highly conserved family member, mechanistic inference from these orthologs to the KT2440 enzyme is well justified.

Key Findings

F001 — PhnX is a Mg^{2+} -dependent phosphonoacetaldehyde hydrolase that cleaves the C–P bond

The core enzymatic activity of PhnX is the hydrolysis of the carbon–phosphorus bond of phosphonoacetaldehyde. The reaction is:



This is an unusual and chemically demanding reaction, because the C–P bond is exceptionally stable — far more resistant to chemical and enzymatic attack than the phosphoester bonds handled by ordinary phosphatases. The enzyme absolutely requires **Mg(II)** as a cofactor. This reaction and cofactor requirement were established biochemically for the *Bacillus cereus* and *Salmonella typhimurium* phosphonatases (PMID: 10956028; PMID: 9649311), which state directly that "phosphonatase catalyzes the hydrolysis of phosphonoacetaldehyde to acetaldehyde and phosphate using Mg(II) as cofactor." The enzyme is classified under **EC 3.11.1.1**, the enzyme class for C–P bond hydrolases. UniProt Q88KT1 (PP_2208) is annotated to the PhnX family of the HAD-like hydrolase superfamily, placing the *P. putida* KT2440 protein squarely in this functional class.

F002 — Catalysis proceeds by a bicovalent mechanism: a Lys53 Schiff base plus an Asp12 phosphoaspartyl intermediate

Phosphonatase does not simply hydrolyze the C–P bond in one step. Instead it uses a remarkable **two-covalent-intermediate ("bicovalent") mechanism**. First, the ϵ -amino group of **Lys53** condenses with the aldehyde carbonyl of phosphonoacetaldehyde to form a **Schiff base (imine)**. This covalent linkage transforms the substrate into an electron sink that labilizes

the adjacent C–P bond. The phosphoryl group is then abstracted and transferred to an **active-site aspartate** (Asp12 in the *B. cereus* enzyme; Asp11 in *S. typhimurium*), producing a **phosphoaspartyl (acyl-phosphate) intermediate**. Finally, water hydrolyzes this phosphoaspartyl intermediate, releasing inorganic phosphate with **retention of configuration at phosphorus** (PMID: 9649311; PMID: 10956028). The mechanism was described as one "in which an active-site nucleophile abstracts the phosphoryl group from the Schiff-base intermediate formed from Lys53 and phosphonoacetaldehyde."

The existence of a genuine covalent catalytic intermediate was demonstrated directly using radiolabeled phosphonoacetaldehyde: single-turnover experiments captured "a kinetically competent covalent intermediate" (PMID: 14596832). Site-directed mutagenesis further validated the mechanism — mutation of Lys53 abolishes activity, and His-56 and Met-49 were shown to assist Schiff-base formation, with Ala substitution of His-56 causing a ~1,000-fold and Leu substitution of Met-49 a ~17,000-fold drop in k_{cat}/K_m (PMID: 14670958). (A QM/MM study, PMID: 18802516, proposed a proton-transfer variant, but the covalent Schiff-base mechanism is supported by direct isotope-trapping evidence.)

F003 — PhnX acts in the two-step 2-AEP degradation pathway, downstream of the aminotransferase PhnW

PhnX does not act on 2-aminoethylphosphonate directly. It operates as the second enzyme of a defined two-step catabolic pathway. In the first step, the aminotransferase **PhnW** converts 2-AEP into phosphonoacetaldehyde (PAA); in the second step, **PhnX cleaves PAA** to acetaldehyde and phosphate (PMID: 33830741). This "phosphonatase pathway acts on the natural Pn alpha-aminoethylphosphonate (AEPn). In a two-step process it leads to cleavage of the C-P bond by a hydrolysis reaction requiring an adjacent carbonyl group" (PMID: 7765831). The pathway is widespread and ecologically significant: roughly **40% of sequenced bacterial genomes encode one or more phosphonate catabolic pathways** (PMID: 22303297).

F004 — Phosphonatase is a cytoplasmic homodimer with a HAD core domain and a mobile cap domain; the active site lies at their interface

The structure of the *B. cereus* phosphonatase reveals a **homodimer**. Each monomer contains an **α/β HAD core domain** built around a centrally located six-stranded parallel β -sheet flanked by α -helices, plus a small **five-helix-bundle "cap" domain** (approximately residues 21–99). The active site is formed **at the interface between the core and cap domains**: the Schiff-base-forming Lys53 resides on the cap domain, while the catalytic Mg(II) and the phosphate/tungstate-binding site (Mg(II) coordinated by Asp12 and Asp186) reside on the core domain

([PMID: 10956028](#)). Because the enzyme carries **no signal or secretion sequence** and acts on an intracellularly generated substrate (phosphonoacetaldehyde produced by cytoplasmic PhnW), PhnX is a **soluble cytoplasmic enzyme** — it performs its function inside the cell.

F005 — PhnX has narrow substrate specificity: it requires an aldehyde/adjacent carbonyl

The Schiff-base mechanism imposes a strict chemical constraint on the substrate. Because the first catalytic step requires condensation of Lys53 with a carbonyl, PhnX can only cleave phosphonates that carry an **adjacent aldehyde group** — phosphonoacetaldehyde being the physiological substrate. This is a defining contrast with the alternative **C–P lyase** route, which has broad substrate tolerance and cleaves both substituted and unsubstituted phosphonates (including glyphosate) via radical/redox chemistry ([PMID: 7765831](#)). The narrow specificity is therefore a direct mechanistic consequence: without the substrate carbonyl the imine cannot form ([PMID: 10956028](#)). Phosphonate catabolic operons are frequently, though not universally, part of the **Pho regulon** under phosphate-starvation control ([PMID: 7946467](#); [PMID: 22303297](#)).

F006 — In *P. putida*, the *phnWX* system enables 2-AEP use as C, N or P source, under dual global + substrate-specific control

This finding is directly relevant to *P. putida* and distinguishes its regulation from the textbook Pho-regulon model. In *P. putida* BIRD-1, **2-AEP can serve as a source of carbon, nitrogen, or phosphorus**. Utilization occurs only upon depletion of the corresponding nutrient — carbon limitation acting through **CbrAB**, nitrogen limitation through **NtrBC**, and phosphorus limitation through **PhoBR**. In addition, full expression requires the substrate itself: the system is **substrate-inducible** via a **LysR-type regulator, AepR**, encoded upstream of the *phnWX* transaminase-phosphonatase genes ([PMID: 35229442](#)). This dual-regulation architecture is corroborated by earlier work on *P. putida* NG2, where both AEP:pyruvate aminotransferase and phosphonatase activities were "inducible by the presence of 2-aminoethylphosphonic acid in the culture medium, **regardless of the phosphate status of the cells**" ([PMID: 9841125](#)) — i.e., not strictly Pho-controlled in *Pseudomonas*.

F007 — The HAD cap domain is the evolutionary innovation that enables C–P cleavage; cap–core closure gates catalysis

Phosphonatase illustrates how the HAD superfamily — dominated by phosphatases — evolved a new reaction. The catalytic innovation is the **C1-type cap domain** inserted into the HAD core; this cap carries the Schiff-base Lys53 and drives **active-site desolvation and catalysis** ([PMID:](#)

[17654544](#); [PMID: 16889794](#)). Catalysis is **conformationally gated**: the enzyme must transition from an open to a closed state, bringing cap and core together. A Lys53Arg mutant structure shifts the equilibrium toward the open conformation, demonstrating that "proton dissociation from the cap domain Lys53 is required for cap domain–core domain closure," and that a **His56–water pair relays a proton to the Pald carbonyl** to enable Schiff-base formation ([PMID: 17070898](#)). A stringently conserved cap-loop glycine (Gly50) positions Lys53 correctly; a Gly50Pro mutation rotates "Lys53, the Schiff Base forming lysine, ... out of the catalytic site" and abolishes activity ([PMID: 15005616](#)).

F008 — PhnX's substrate is supplied by PhnW, a PLP-dependent 2-AEP:pyruvate aminotransferase

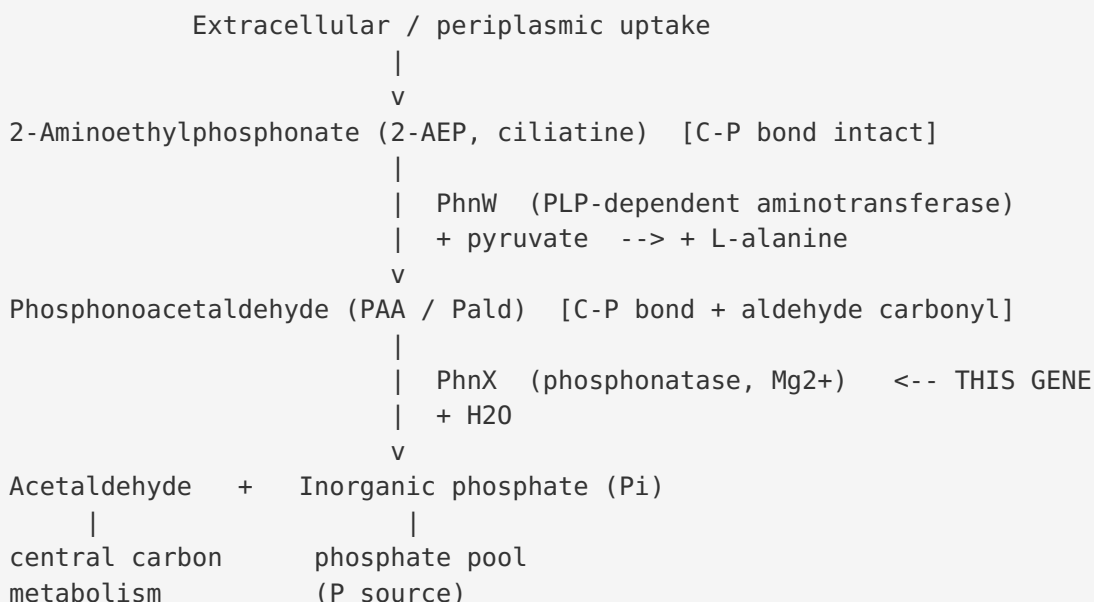
The upstream partner enzyme, **PhnW** (AEP transaminase, AEPT), is a **pyridoxal-5'-phosphate (PLP)-dependent type I aminotransferase**. It "catalyzes the transamination of 2-aminoethylphosphonate (AEP) with pyruvate to phosphonoacetaldehyde and l-alanine" ([PMID: 33743347](#)), thereby producing exactly the substrate that PhnX consumes. A 2.2 Å crystal structure shows a two-domain type-I aminotransferase fold with bound PLP and the product phosphonoacetaldehyde, defining substrate recognition and stereospecific α -proton elimination ([PMID: 12403617](#)). Notably, AEPT is "an enzyme essential to phosphonate synthesis and degradation pathways" — the same transamination chemistry runs in reverse during phosphonate biosynthesis. The tight metabolic coupling of PhnW → PhnX explains why the two genes are co-organized and co-regulated as the *phnWX* unit.

F009 — Biological significance: PhnX-mediated 2-AEP catabolism is a nutrient-scavenging adaptation

Phosphonates "contain a direct C–P bond that is particularly resistant to chemical and enzymatic degradation," yet they are environmentally ubiquitous. Bacteria in phosphate-limited environments "have evolved systems to uptake and catabolize phosphonates" as a nutrient resource ([PMID: 37836707](#)). The hydrolytic phosphonatase (PhnW→PhnX) route is one of **at least three mechanistically distinct C–P-cleaving strategies** — hydrolytic, radical (C–P lyase), and oxidative ([PMID: 37836707](#); [PMID: 7765831](#)). Given that ~40% of sequenced bacterial genomes carry phosphonate catabolic genes ([PMID: 22303297](#)), PhnX-type enzymes contribute materially to global phosphorus cycling. In *P. putida*, a versatile soil saprophyte, this system contributes to nutritional flexibility by unlocking ciliatine (2-AEP), one of the most abundant biogenic phosphonates.

Mechanistic Model and Interpretation

The complete pathway and the place of PhnX



The pathway **partitions the three elements** of 2-AEP into usable pools: the amino nitrogen is released by PhnW (as the transamination product), the phosphorus is released by PhnX as inorganic phosphate, and the carbon skeleton exits as acetaldehyde, which feeds central metabolism. This is why 2-AEP can serve independently as a C, N, or P source depending on which nutrient the cell lacks.

The catalytic cycle of PhnX (bicovalent mechanism)

- Step 1 Cap-core OPEN --> CLOSED (gated by Lys53 deprotonation; conserved Gly50 positions Lys53)
- Step 2 Lys53-NH₂ + O=CH-CH₂-P₀₃(2-) --> Lys53-N=CH-CH₂-P₀₃ (SCHIFF BASE)
(His56-water relays proton to carbonyl)
- Step 3 C-P bond cleavage; phosphoryl transferred to Asp12 --> Asp12-O-P₀₃ (PHOSPHOASPARTYL)
+ enamine/acetaldehyde released
- Step 4 H₂O hydrolyzes phosphoaspartyl intermediate --> Pi released (retention at P)
- Step 5 Cap-core reopens; enzyme regenerated

Two features make this enzyme special. First, the **Schiff base** converts an otherwise inert phosphonate into a reactive species — the imine's electron-withdrawing character labilizes the C-P bond, a trick no ordinary phosphatase needs. Second, the **cap domain** is the physical

embodiment of this innovation: it delivers Lys53 into the active site and, by closing over the core, excludes bulk water so that the covalent chemistry can proceed. The HAD core, meanwhile, contributes the conserved Mg^{2+} /aspartate machinery for the phosphoryl-transfer and phosphoaspartyl-hydrolysis steps that are shared with the rest of the superfamily.

Comparison of bacterial C-P bond-cleaving strategies

Feature	Phosphonatase (PhnX)	C-P lyase (PhnGHIJKLM)	Phosphonoacetate hydrolase
Mechanism	Hydrolytic, Schiff-base + phosphoaspartyl	Radical (SAM-dependent)	Hydrolytic
Substrate scope	Narrow — requires aldehyde (phosphonoacetaldehyde)	Broad — substituted & unsubstituted phosphonates	Phosphonoacetate
Metal/cofactor	Mg^{2+}	Fe-S, SAM, ATP	Zn^{2+} (typical)
Physiological substrate	2-AEP (via PhnW)	e.g., methylphosphonate, glyphosate	Phosphonoacetate
In <i>P. putida</i> KT2440	Present (phnWX)	Route present in many bacteria	—

Regulation in *P. putida* (integrated logic)

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Carbon starvation --> CbrAB --\
Nitrogen starvation --> NtrBC ----> permit phnWX expression
Phosphate starvation--> PhoBR --/          AND
Presence of 2-AEP --> AepR (LysR) --> substrate induction (required for full expression)
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The *P. putida* system is therefore an **AND gate**: the cell only invests in the *phnWX* machinery when (a) it is short of C, N, or P, **and** (b) the substrate 2-AEP is actually available. This avoids wasteful expression and distinguishes *Pseudomonas* from organisms where phosphonate catabolism is governed purely by phosphate starvation (Pho regulon).

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
10956028	<i>B. cereus</i> phosphonatase crystal structure	Defines reaction + Mg(II) cofactor; homodimer; HAD core + cap domain; active site at interface; bicovalent mechanism (F001, F002, F004, F005)
9649311	Mechanism from sequence analysis & mutagenesis	Confirms reaction/products; proposes Schiff base (Lys53) + phosphoaspartyl (Asp11/12) steps (F001, F002)
14596832	Radiolabeled phosphonoacetaldehyde synthesis	Direct evidence of a kinetically competent covalent intermediate (F002)
14670958	X-ray + mutagenesis of Schiff-base formation	His-56, Met-49 assist Schiff base; cap-core association positions Lys53 (F002, F007)
33830741	Recurrent enzyme in phosphonate degradation	Defines two-step PhnW→PhnX pathway and PhnX's role (F003)
7765831	Molecular genetics of C–P cleavage (review)	Narrow specificity, adjacent-carbonyl requirement; contrast with C–P lyase (F003, F005, F009)
22303297	Genes/enzymes of phosphonate metabolism	~40% of genomes encode phosphonate catabolism; regulation modes (F003, F005, F009)
7946467	Microbial S/P xenobiotic metabolism	Pho-regulon control of phosphonate genes (F005)
35229442	2-AEP utilization in <i>P. putida</i> BIRD-1	Dual global (CbrAB/NtrBC/PhoBR) + AepR substrate induction; 2-AEP as C/N/P source (F006)
9841125	Phosphate-independent 2-AEP degradation in <i>P. putida</i> NG2	Substrate-inducible, phosphate-independent activities (F006)
17654544	Pseudomonas-specific HAD subfamily	Cap domain role in desolvation/catalysis (F007)
16889794	Evolutionary genomics of the HAD superfamily	C1/C2 cap insertions drove functional diversification; squiggle/flap open–closed motion (F007)
17070898		

PMID	Title (abbrev.)	How it supports the findings
	Cap domain in hydrolytic P–C cleavage	Open/closed gating; Lys53 deprotonation → domain closure; His56–water proton relay (F007)
15005616	Substrate-specificity loop of HAD cap	Conserved Gly positions Lys53; Gly→Pro rotates Lys53 out, abolishing activity (F007)
33743347	AEP:pyruvate aminotransferase (<i>P. aeruginosa</i>)	PhnW = PLP-dependent transaminase producing PAA + L-alanine (F008)
12403617	Crystal structure of AEP transaminase	Structure of PhnW/AEPT; essential to synthesis & degradation (F008)
37836707	Microbial degradation of phosphonates (review)	Ecological driver; three mechanistic classes of C–P cleavage (F009)
18802516	QM/MM of phosphonatase	Proposes proton-transfer variant of mechanism (alternative view; challenges covalent model)

Consistency of the evidence. Multiple independent lines — X-ray crystallography, site-directed mutagenesis, isotope trapping, gene-sequence analysis, and physiological/genetic studies in *P. putida* — converge on the same functional model. The one dissenting mechanistic proposal (PMID: [18802516](#)) is a computational QM/MM study suggesting a proton-transfer route rather than a discrete covalent intermediate; however, the direct isotope-trapping of a kinetically competent covalent species (PMID: [14596832](#)) provides strong experimental support for the Schiff-base/phosphoaspartyl model.

Limitations and Knowledge Gaps

- 1. No direct biochemical or structural characterization of the KT2440 enzyme itself.** All mechanistic and structural detail derives from orthologs — chiefly *Bacillus cereus* and *Salmonella typhimurium* phosphonatases. Because PhnX is a highly conserved, HAMAP-rule-annotated family member, inference to Q88KT1 is well justified, but the precise kinetic parameters (k_{cat} , K_m , metal preference) of the *P. putida* KT2440 PhnX have not been measured directly in the reviewed literature.

2. **Residue numbering must be mapped.** Catalytic residues (Lys53, Asp12, Asp186, His56, Met49, Gly50) are numbered for the *B. cereus* enzyme. The exact positions in the KT2440 sequence (Q88KT1) require an explicit sequence alignment to confirm (not performed in these iterations).
3. **Regulatory data come from *P. putida* BIRD-1 and NG2, not KT2440.** While KT2440 is expected to share the *aepR/phnWX* architecture, the specific promoter/operator arrangement and the confirmation of AepR function in KT2440 were not directly established here.
4. **Uptake and downstream steps unresolved.** How 2-AEP is transported into the KT2440 cytoplasm, and the fate/regulation of acetaldehyde disposal, were not investigated in detail.
5. **Localization is inferred, not experimentally demonstrated for KT2440.** The cytoplasmic assignment rests on the absence of a signal sequence and on the intracellular origin of the substrate, consistent with the soluble structures of orthologs — but no proteomic localization data specific to KT2440 were reviewed.

Proposed Follow-up Experiments / Actions

1. **Sequence-align Q88KT1 to *B. cereus*/*S. typhimurium* phosphonatases** to confirm conservation and exact positions of Lys53, Asp12, Asp186, His56, Met49, and the cap-loop Gly — verifying the catalytic machinery is intact in KT2440.
2. **Recombinantly express and purify KT2440 PhnX (PP_2208)** and measure steady-state kinetics (k_{cat} , K_m) on phosphonoacetaldehyde, plus Mg^{2+} dependence, to obtain species-specific parameters.
3. **Reconstitute the two-enzyme PhnW→PhnX cascade in vitro** from KT2440 proteins to confirm coupled turnover of $2\text{-AEP} \rightarrow \text{acetaldehyde} + \text{Pi} + \text{L-alanine}$.
4. **Test substrate specificity** with a panel of phosphonates (with and without adjacent carbonyls; e.g., 2-AEP vs. methylphosphonate) to experimentally confirm the aldehyde requirement predicted by the Schiff-base mechanism.
5. **Genetic/regulatory validation in KT2440:** construct $\Delta phnX$, $\Delta phnW$, and $\Delta aepR$ mutants and assay growth on 2-AEP as sole C, N, or P source; use reporter fusions to confirm CbrAB/NtrBC/PhoBR + AepR control in KT2440 specifically.

6. **Determine or model the KT2440 PhnX structure** (crystallography or AlphaFold + validation) to confirm the homodimer, HAD core, and C1 cap architecture and the open/closed conformational states.
 7. **Localization confirmation:** subcellular fractionation or fluorescent tagging to verify cytoplasmic localization in KT2440.
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Conclusion

phnX (PP_2208; Q88KT1) in *Pseudomonas putida* KT2440 encodes **phosphonoacetaldehyde hydrolase (phosphonatase, EC 3.11.1.1)**, a cytoplasmic, Mg^{2+} -dependent homodimeric enzyme of the HAD superfamily. It catalyzes the hydrolytic cleavage of the carbon–phosphorus bond of phosphonoacetaldehyde to yield acetaldehyde and inorganic phosphate, using a distinctive bicovalent mechanism (a Lys53 Schiff base followed by an Asp12 phosphoaspartyl intermediate) enabled by an evolutionarily innovative cap domain and gated by cap–core conformational closure. It performs the second, committed step of the two-enzyme *phnWX* 2-aminoethylphosphonate degradation pathway — downstream of the PLP-dependent aminotransferase PhnW — allowing *P. putida* to exploit the abundant biogenic phosphonate ciliate as a source of phosphorus, nitrogen, and/or carbon. In *P. putida* the pathway is substrate-inducible through the LysR-type regulator AepR, integrated with the global CbrAB/NtrBC/PhoBR nutrient-stress networks, rather than being strictly phosphate-starvation controlled.
